

AYMANE AIT DADS

Machine Learning Intern, Applied Mobility Intelligence | Sensor Data Fusion · Predictive Modeling · ML Pipeline Engineering

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PROFESSIONAL SUMMARY

Ranked **17th out of 3,000+ teams** in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by designing and executing an end-to-end LoRA SFT pipeline on a 30B-parameter model with BF16 precision, staged continuation training, and controlled ablation experiments. At Orange Maroc, built Random Forest and K-Means models on **100,000+ fiber-optic network records**, reducing manual analysis time by ~60% and delivering actionable maintenance recommendations directly to the network optimization team. Combines strong foundations in PyTorch, Scikit-learn, feature engineering, and ensemble methods with hands-on experience building production-grade ML pipelines that translate raw sensor and behavioral data into structured, decision-ready outputs - precisely the skill set CMT applies to IoT and telematics data. Aims to contribute applied ML research and engineering depth to CMT's DriveWell platform, advancing driver behavior modeling and predictive analytics across connected vehicle and mobility datasets.

CORE COMPETENCIES

Predictive Modeling | Sensor Data Analysis | Feature Engineering | Ensemble Methods | Anomaly Detection | ML Pipeline Development | Behavioral Pattern Recognition | Data-Driven Insights

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built **Random Forest and K-Means predictive models** on 100,000+ fiber-optic network records - including upload speed, latency, and jitter signals - to classify network quality across thousands of nodes.
- Developed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure planning.
- Reduced manual analysis time by **~60%** through automated feature engineering, pipeline orchestration, and systematic model evaluation across large-scale network datasets.
- Delivered stakeholder presentations translating **ML model outputs** into actionable maintenance recommendations, bridging data science findings and operational decision-making.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge (Kaggle - In Progress)

Kaggle Competition · 2025-2026 · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context), targeting all attention projections, MLP layers, and lm_head with grad accumulation of 32.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers to produce 605 verified cipher rows and 64 verified bit-manipulation rows for clean CoT training data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton · Kaggle GPU

Anomalous Sound Detection - Industrial Equipment

EURECOM Course Project · 2025 · AUC 0.80

- Designed an unsupervised **anomaly detection** system for slide-rail industrial machinery using a Transformer autoencoder trained on Mel spectrograms with SpecAugment data augmentation, achieving **AUC 0.80** for predictive maintenance without labeled anomaly examples.
- Engineered the full ML pipeline from raw audio signal preprocessing through spectrogram feature extraction to model evaluation, demonstrating sensor data analysis skills directly applicable to IoT and telematics behavioral pattern recognition.

PyTorch · Mel Spectrograms · SpecAugment · Transformer Autoencoder

AI-Generated Image Detection - NTIRE 2026 @ CVPR

EURECOM + NTIRE 2026 @ CVPR CodaBench · 2025 · 1st in Class (0.791 AUC)

- Ranked **1st in EURECOM class** on a private Kaggle leaderboard (0.791 AUC) and submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark, fine-tuning a **CLIP ViT-L/14 (428M parameters)** backbone with a custom MLP head and dual learning rates (2e-6 for CLIP, 5e-4 for head) across 250,000 training images.
- Built a **3-model ensemble** weighted by validation AUC with 10-view TTA (flips, crops, blur, resizes), discovering that 1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC) - a key finding on out-of-distribution generalization critical for real-world deployment.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · AdamW · Kaggle

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | Hugging Face Transformers | Unsloth / TRL / PEFT | LoRA SFT | vLLM | BF16/FP16 Mixed Precision

Data Science & Modeling: Python | Pandas | NumPy | Feature Engineering | Ensemble Methods | Clustering (K-Means) | Anomaly Detection | Power BI

Computer Vision & NLP: CLIP | ViT | CNN | DenseNet | Mel Spectrogram Analysis | Transformer Fine-Tuning | Sentiment Analysis | Chain-of-Thought Reasoning